

# Digital Twin-Enhanced Reinforcement Learning for Intelligent xApps Management in O-RAN Systems

Zhizhou He <sup>ID</sup>, *Member, IEEE*, Ahmed Al-Tahmeesschi <sup>ID</sup>, *Member, IEEE*, Chuan Heng Foh <sup>ID</sup>, *Senior Member, IEEE*, Hamed Ahmadi <sup>ID</sup>, *Senior Member, IEEE*, and Mohammad Shojafar <sup>ID</sup>, *Senior Member, IEEE*

## ABSTRACT

Open Radio Access Network (O-RAN) introduces xApps to enable flexible, near real-time control of RAN functions. However, managing multiple xApps dynamically, under rapidly changing network conditions, becomes significantly more challenging in IoT-driven environments, where massive numbers of low-power devices may simultaneously access the network, triggering phenomena such as wake-up storms. In this work, we showcase and compare Digital Twin-enhanced Multi-Agent Reinforcement Learning (MARL) strategies for intelligent xApp management in IoT-enabled O-RAN systems. We explore how Digital Twins can simulate large-scale, event-driven IoT device behaviors, enabling proactive optimization of xApp scheduling while reducing communication and synchronization overhead. We further examine advanced MARL approaches, such as Hierarchical Agent Policy Optimization (HAPPO), to address scalability and coordination challenges under ultra-dense, low-latency, and energy-constrained LPWAN protocols like NB-IoT and LTE-M. We define two system-level KPIs to evaluate the performance of RL schemes: Conflict Avoidance Efficiency and Coordination QoS Improvement. The proposed MARL-based approach significantly outperforms conventional single-agent RL methods in both coordination effectiveness and overall QoS delivery, while maintaining a reasonable model complexity suitable for practical deployment.

## I. INTRODUCTION

### A. MOTIVATION AND BACKGROUND

The evolution toward Open Radio Access Networks (O-RAN) aims to enable vendor-neutral, flexible, and intelligent management of radio access infrastructures [1], [2]. A central innovation in O-RAN is the introduction of xApps running on the Near-Real-Time RAN Intelligent Controller (Near-RT RIC), providing modularized control such as resource allocation, interference management, and mobility optimization [3].

However, emerging Internet of Things (IoT) environments are radically reshaping the operational landscape of O-RAN systems [4]. Unlike traditional human-centric mobile networks characterized by

high data rates and mobility, IoT networks prioritize massive connectivity, low-power operation, and ultra-reliable low-latency communication (URLLC). The applications of O-RAN IoT are showcased in Fig. 1. The proliferation of low-power wide-area network (LPWAN) technologies, such as NB-IoT and LTE-M, further emphasizes the need for efficient management of thousands to millions of static or semi-static IoT devices [5].

A particular challenge in IoT-driven O-RAN lies in managing “wake-up storm”—scenario where large groups of devices simultaneously attempt to access the network following event triggers, resulting in severe signaling congestion and access failures. Traditional xApp management frameworks, which are often designed around periodic control and traffic-driven optimization, are poorly suited to handling such highly event-driven and bursty behaviors. Moreover, frequent communication and synchronization among independently operating xApps during training introduce significant overhead, further exacerbating scalability and coordination challenges in dense IoT deployments [6].

In this context, Digital Twin (DT) technology offers a promising avenue by creating real-time, high-fidelity virtual replicas of physical networks [7]. By simulating IoT device behaviors and network dynamics, DT can predict potential conflicts among xApps, optimize scheduling decisions, and minimize unnecessary communication overhead. Furthermore, the inherent complexity and distributed nature of IoT-driven RAN management naturally align with Multi-Agent Reinforcement Learning (MARL) approaches, where each xApp can be modeled as an autonomous agent learning to optimize its actions in coordination with others. This paper explores the synergy between DT modeling and MARL strategies for intelligent xApp management in IoT-enabled O-RAN systems, highlighting their potential to address emerging scalability, coordination, and energy-efficiency challenges.

### B. LIMITATIONS OF TRADITIONAL xAPP MANAGEMENT

Traditional xApp management frameworks in O-RAN systems were primarily designed to optimize human-centric traffic patterns (predictable, periodic usage patterns from human behavior, contrasted with bursty,

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The authors are with the University of Surrey, GU2 7XH Guildford, U.K., and also with the University of York, YO10 5DD York, U.K. Zhizhou He is the corresponding author.

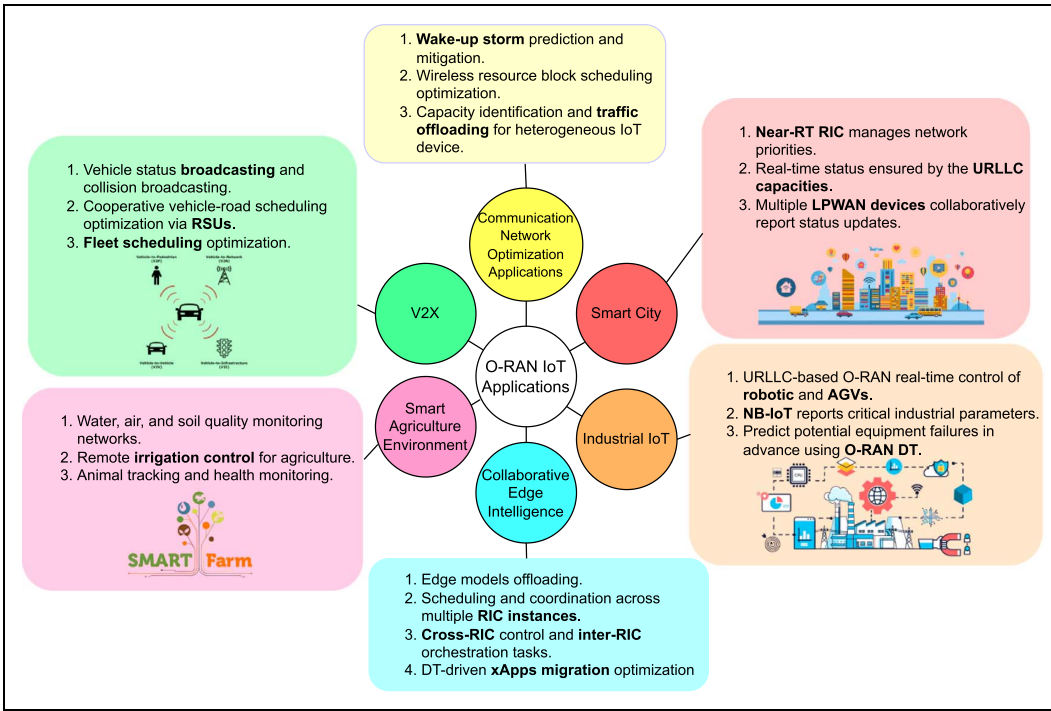


FIG. 1. O-RAN IoT applications.

asynchronous IoT traffic), focusing on improving throughput, handover efficiency, and spectral utilization. These frameworks typically employ rule-based policies or single-agent reinforcement learning (RL) methods, where each xApp independently makes decisions based on local observations or predefined network policies. However, in IoT-driven O-RAN environments, these traditional approaches exhibit several critical limitations:

- **Scalability Issues:** The massive number of IoT devices generates frequent and unpredictable access patterns. Rule-based or isolated xApp optimization cannot scale effectively to manage simultaneous access attempts, leading to congestion and degraded quality of service (QoS).
- **Synchronization Overhead:** Traditional xApps often require frequent information exchange and synchronization through control-plane signaling during AI training. In ultra-dense IoT deployments, this leads to significant signaling overhead, increased latency, and excessive load on the Near-RT RIC.
- **Inflexibility to Event-Driven Behaviors:** Most existing xApp control loops assume periodic monitoring and decision-making cycles. Such designs are ill-suited to highly event-driven IoT behaviors, where access requests may occur in sudden bursts following external triggers (e.g., wake-up storms).
- **Inadequate Conflict Resolution:** Independent xApps may generate conflicting control actions without global awareness, especially under dynamic network conditions. Static conflict resolution mechanisms are insufficient in complex, rapidly evolving IoT scenarios.
- **Energy Inefficiency:** IoT devices often operate under stringent energy constraints. Traditional xApp policies optimized for throughput or latency do not explicitly account for minimizing device energy consumption, leading to suboptimal performance in LPWAN-based IoT systems.

These limitations highlight the urgent need for more intelligent, coordinated, and predictive xApp management mechanisms capable of operating

efficiently in large-scale, IoT-enabled O-RAN environments. Addressing these challenges requires the integration of DT technology for environment simulation and MARL strategies for scalable, adaptive decision-making.

### C. PROPOSED APPROACH: DT + MARL

This article presents a novel investigation into DT-enhanced MARL schemes for intelligent xApp management in IoT-enabled O-RAN systems. The main contributions are summarized as follows:

- We propose a DT-assisted MARL framework tailored for event-driven, large-scale IoT environments. The framework enables proactive xApp scheduling while mitigating synchronization and communication overhead during training. The detailed workflow of DT implementation and its interaction mechanisms is presented in a later section.
- We formulate two coordination-oriented KPIs—Conflict Avoidance Efficiency (CAE) and Coordination QoS Score (CQS)—to quantitatively evaluate inter-xApp interactions and system-level performance under RL control.
- We demonstrate that advanced MARL methods significantly outperform conventional single-agent RL in both CAE and QoS metrics, while maintaining acceptable model complexity suitable for practical deployment.
- We provide a detailed computational complexity analysis (training time, inference latency, and memory footprint), offering insights into the feasibility of deploying each algorithm in real-world RAN infrastructure.

## II. FUNDAMENTALS

### A. O-RAN ARCHITECTURE AND xAPP MANAGEMENT

The O-RAN architecture adopts a disaggregated base station model, most commonly based on the 3GPP NR 7.2 split [8], which separates functionality into the

## B. DIGITAL TWIN MODELING

DT, as a high-fidelity virtual representation of the O-RAN system, offers a powerful sandbox for simulating “what-if” scenarios, identifying potential risks, and validating mitigation strategies before deployment [9]. This proactive capability enables rapid prototyping and intelligent decision-making, which is particularly valuable for complex xApp orchestration and coordination tasks.

At a high level, the DT is positioned between the Near-RT RIC and Non-RT RIC, and its architecture comprises three primary components: (i) the Physical Twin (PT), representing the live RAN network including O-RU, O-DU, and O-CU nodes, which serve as data sources; (ii) the DT model, which hosts dynamic state representations, predictive models, and simulation tools; and (iii) a bi-directional real-time data flow pipeline between the PT and DT, and between DT and RIC entities. This architecture enables continuous learning and real-time emulation, supporting efficient, decoupled xApp coordination and life-cycle optimization.

In the context of xApp management, one of the key advantages of DT is its ability to minimize direct communication and synchronization overhead among xApps. Traditionally, multiple xApps operating in parallel may require frequent state exchange or conflict resolution via control-plane signaling, especially when overlapping control domains or time-sensitive decisions are involved. Such interactions can lead to scalability bottlenecks and increased latency at the Near-RT RIC. By simulating the RAN environment and xApp interactions internally, the DT enables indirect coordination, allowing xApps to evaluate their policies in a shared virtual environment without requiring explicit inter-xApp messaging. This greatly reduces E2 interface signaling, avoids race conditions, and enables more scalable, modular deployment of heterogeneous xApps. It is worth noting that although the DT involves bi-directional data exchange with the Physical Twin, the potential runtime overhead is mitigated by design. Specifically, the majority of the MARL agent training occurs offline within the DT simulation, using pre-collected or synthetic datasets, thereby avoiding unnecessary burden on the live system. During online inference, only lightweight agent outputs—such as action vectors and scalar rewards are exchanged between the DT and xApps. This efficient data handling ensures the DT-enhanced architecture remains practical for real-world deployment without introducing excessive latency or signaling cost.

The overall control workflow in our proposed DT-enhanced RL framework spans multiple layers of the O-RAN architecture, including the Non-RT RIC, DT, Near-RT RIC, E2 nodes, beamforming units, and IoT UEs, as shown in Fig. 3. The process begins with the construction of a high-fidelity DT model that emulates RAN dynamics under IoT traffic scenarios. Within this environment, we conduct offline training of multiple MARL agents, each designed to learn a control policy targeting a specific RAN optimization task such as transmission power control or PRB-based bandwidth allocation. These trained policies are then exported and encapsulated as xApps, which are deployed in the Near-RT RIC.

At runtime, candidate actions proposed by the deployed xApps are first evaluated within the DT for estimated CAE and QoS outcomes. This enables real-

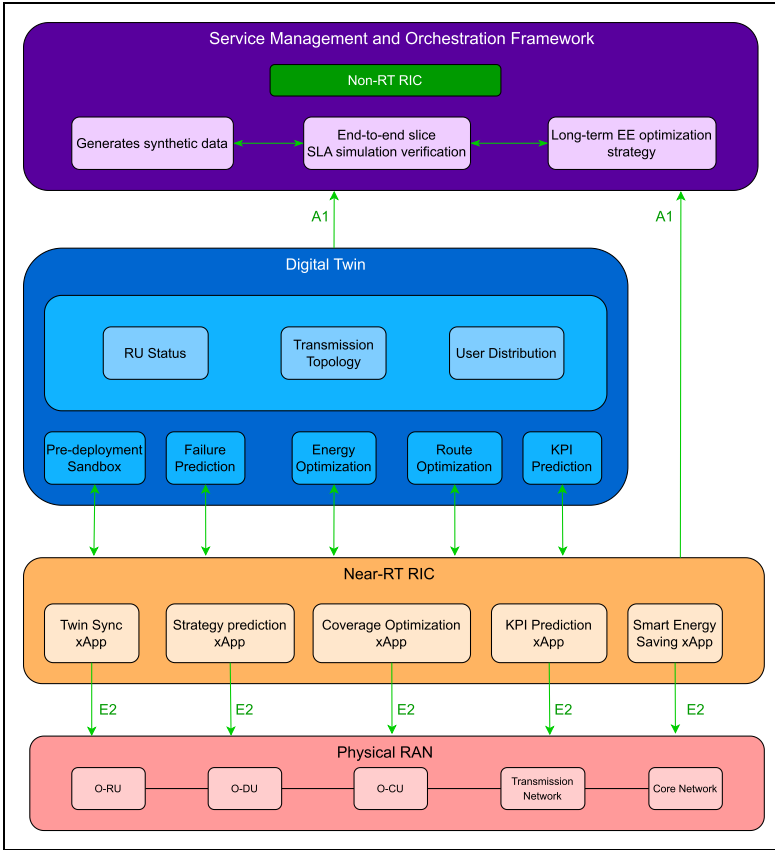


FIG. 2. Digital twin-based O-RAN architecture.

Central Unit (CU), Distributed Unit (DU), and Radio Unit (RU), as shown in Fig. 2. To emphasize openness and interoperability, O-RAN denotes these components as O-CU, O-DU, and O-RU<sup>1</sup>. The O-CU is further divided into the control and user planes, enabling modular and flexible deployment. A key innovation in O-RAN is the introduction of the RAN Intelligent Controller (RIC), which brings intelligence and programmability to the RAN via standardized interfaces. The RIC is composed of two layers: the Near-RT RIC, operating on 10 ms to 1 s time scales, and the Non-Real-Time RIC (Non-RT RIC), operating on scales above 1 s and integrated within the Service Management and Orchestration (SMO) framework.

Within this architecture, modular applications known as xApps (for Near-RT RIC) and rApps (for Non-RT RIC) execute control logic such as load balancing, handover optimization, interference mitigation, and admission control. xApps interact with underlying RAN components (O-CU and O-DU) through the standardized E2 interface, and may coordinate with rApps via the A1 interface. These applications subscribe to and process Key Performance Indicators (KPIs), such as throughput, delay, or resource block usage, to monitor network state and issue timely control actions. However, as the number of deployed xApps grows, especially in heterogeneous and dynamic networks, managing their interaction, conflict resolution, and life-cycle becomes increasingly complex. Traditional management relies on static policies or isolated logic, which struggle to scale with growing deployment density and performance demands. This motivates the exploration of intelligent, coordinated xApp management frameworks, discussed in the following sections.

<sup>1</sup> The prefix “O-” indicates compliance with O-RAN-defined open interfaces and reference architecture.



time policy filtering to prevent coordination failures. Approved actions are subsequently executed through RL actor inference and applied to E2 and reconfigurable intelligent surface (RIS) nodes for adaptive scheduling, beamforming, and resource management. Key performance indicators (KPIs) from the RAN are continuously fed back into the DT for periodic synchronization and lightweight fine-tuning. This end-to-end workflow supports proactive, closed-loop optimization of xApp behavior under dense, dynamic, and resource-constrained IoT environments.

Furthermore, the RL agents in this paper is trained in an offline reinforcement learning fashion, where policies are trained entirely using previously collected or synthetic datasets within the DT environment. This avoids risky or disruptive exploration in the live RAN system. Offline RL enables stable training cycles, supports scenario replay (e.g., wake-up storms), and allows seamless bootstrapping of policies before online deployment. Later fine-tuning is conducted with limited real-time feedback to improve adaptation while preserving system safety.

### C. REINFORCEMENT LEARNING AND MULTI-AGENT REINFORCEMENT LEARNING

1) *Traditional RL: Advantages and Limitations:* In the context of O-RAN xApp management, traditional RL techniques have been applied to tasks such as dynamic spectrum allocation, handover optimization, and user scheduling. A typical formulation models the environment as a Markov Decision Process (MDP), where the agent—often implemented as an xApp—observes the current network state, selects an action such as adjusting resource block allocation, and receives a reward based on network KPIs.

Traditional RL methods, particularly model-free algorithms such as Deep Q-Networks (DQN), Proximal Policy Optimization (PPO), provide several advantages in this setting. These methods allow RL agents to autonomously adapt and learn optimal control strategies directly through interaction with the environment, without relying on predefined system models. Moreover, they are capable of optimizing long-term cumulative rewards, enabling xApps to account for both immediate and delayed consequences of control actions. Another appealing feature of these methods is their model-agnostic nature, which allows for flexible application across various RAN use cases without extensive domain-specific customization.

Despite these strengths, traditional RL approaches face significant limitations when applied to the inherently distributed and dynamic nature of multi-xApp O-RAN environments. Reinforcement learning can be categorized into single-agent (SA-RL) and multi-agent paradigms, depending on whether the environment includes other learning agents. Traditional single-agent RL typically assumes a stationary environment, where the transition dynamics do not change over time. However, in O-RAN settings with multiple concurrently operating xApps, the environment becomes non-stationary from each agent's perspective, which necessitates the use of MARL techniques. In contrast, O-RAN systems often involve multiple xApps simultaneously issuing control actions, which creates a non-stationary and partially observable environment for each agent. Second, the scalability of RL frameworks becomes a bottleneck as the number of xApps and associated control variables increases. The exponential

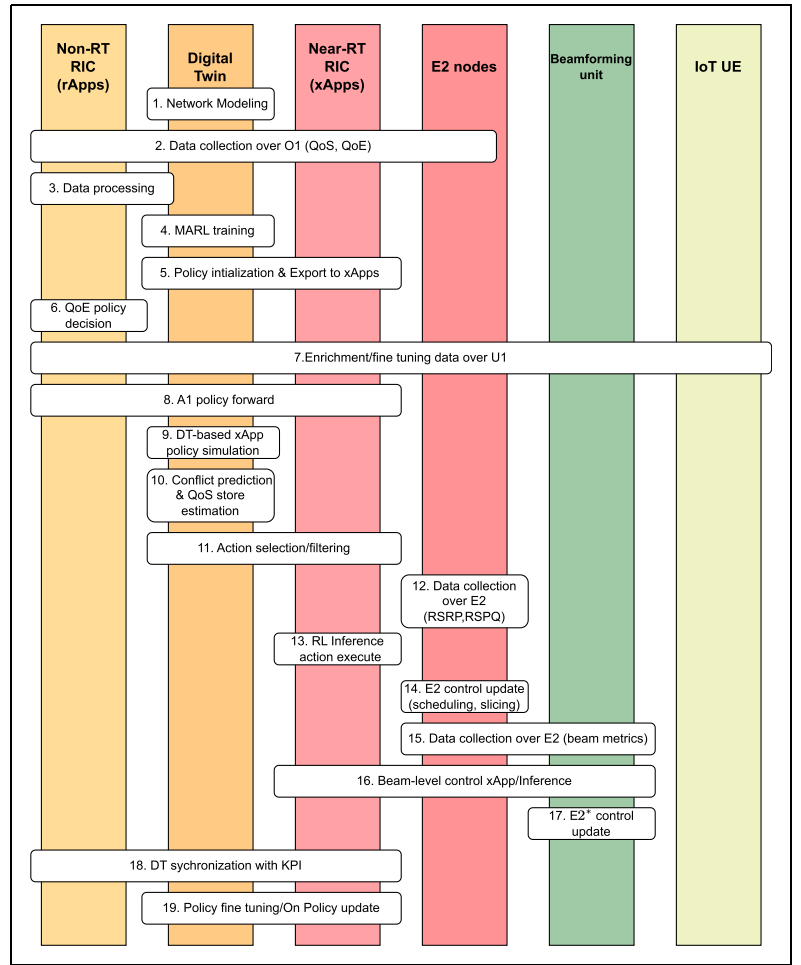


FIG. 3. Digital twin-based O-RAN architecture workflow.

growth of the state and action space leads to slower convergence and reduced training stability. Third, independently trained RL agents lack explicit coordination mechanisms, making it difficult to avoid conflicting or redundant decisions. This can result in performance degradation or even instability at the RAN layer. Finally, the high sample complexity of RL methods imposes a substantial burden when applied to live networks, where extensive exploration may compromise service quality or violate operational constraints.

2) *Advanced MARL:* The limitations raised by traditional RL underscore the necessity of a more robust and coordinated learning framework. MARL extends traditional RL by enabling multiple decentralized agents, such as distributed xApps, to collaboratively learn policies that optimize global network performance while preserving local autonomy and scalability. Centralized Training with Decentralized Execution (CTDE) is a widely adopted learning paradigm in MARL. Under CTDE, agents are trained using global information—such as the joint state or other agents' actions to improve learning stability and coordination. However, during execution, each agent acts independently based only on its local observations. This architecture strikes a balance between collaborative learning and distributed decision-making, making it well-suited for O-RAN environments where xApps must operate autonomously yet cooperatively.

In this section, we going to examine five types of advanced and scalable MARL algorithms that are

|                        | DQL   | PPO    | MAPPO              | HAPPO               | FACMAC          | CommNet              | EC-RL              |
|------------------------|-------|--------|--------------------|---------------------|-----------------|----------------------|--------------------|
| Training Stability     | Low   | High   | High               | Very High           | High            | Medium               | Medium             |
| Scalability            | Low   | Medium | High               | High                | Medium          | Low                  | Medium             |
| Communication Modeling | No    | No     | No                 | No                  | Implicit        | Explicit             | Explicit           |
| CTDE                   | No    | No     | Yes                | Yes                 | Yes             | Partial              | No                 |
| Coordination Mechanism | None  | None   | Centralized Critic | Hierarchical Update | Attention-based | Shared Communication | Emergent protocols |
| Agent                  | SA-RL | SA-RL  | MARL               | MARL                | MARL            | MARL                 | MARL               |

FIG. 4. A comparison between various reinforcement learning schemes.

suitable for O-RAN xApps management, summarized as in Fig. 4.

- Multi-Agent PPO (MAPPO). MAPPO extends the widely used PPO algorithm to the multi-agent setting under the CTDE paradigm [10]. Each agent learns its own policy using a shared or individual value function while considering other agents' behavior during training. MAPPO is known for its training stability, robustness, and strong empirical performance across a wide range of cooperative tasks. Its simplicity and scalability make it well-suited for managing independent xApps with partial observability in O-RAN environments.
- Hierarchical Agent PPO (HAPPO). HAPPO builds upon MAPPO by introducing a hierarchical training structure to improve sample efficiency and scalability [11]. It updates policies sequentially in a way that considers how one agent's policy update affects others, which is crucial in tightly coupled systems. In the context of O-RAN, HAPPO is especially useful for coordinating large numbers of interdependent xApps while maintaining stable learning across overlapping control domains.
- Factored Multi-Agent Centralised policy gradients (FACMAC). FACMAC introduces an attention-based centralized critic to model inter-agent dependencies more effectively [12]. The attention mechanism allows each agent to selectively focus on relevant information from other agents during training, improving credit assignment and coordination. FACMAC is particularly beneficial in heterogeneous or dynamic xApp environments where different agents (xApps) may need to cooperate flexibly without explicit communication protocols.
- CommNet. CommNet is one of the earliest neural architectures designed for differentiable inter-agent communication [13]. It enables agents to share internal hidden states during training and execution through a shared communication channel. While relatively simple, CommNet demonstrates how explicit communication among agents can enhance coordination. In O-RAN, it can serve as a conceptual foundation for exploring lightweight, communication-efficient xApp collaboration mechanisms.

- Emergent Communication (EC) MARL [14] refers to the ability of agents in a multi-agent system to autonomously develop communication protocols through interaction, without being explicitly programmed with a predefined language. These protocols emerge during training as a byproduct of coordination under shared objectives. EC enables agents to learn when, what, and how to communicate based on task-driven reward feedback and environmental signals. In contrast to DT-based coordination, which relies on offline simulation and supervision, EC operates in an online fashion and supports decentralized, real-time policy negotiation among agents. To sum up, EC MARL employs explicit, token-based message channels to enable agents to learn coordination protocols during training, offering moderate scalability and training stability. While it supports flexible multi-agent interactions without centralized critics, its convergence is sensitive to reward shaping and becomes challenging in large-scale deployments.

### III. PERFORMANCE EVALUATION

#### A. SIMULATION SETUP

We utilize a synthetic UE-level dataset collected from a Viavi RIC testbed, containing 25 features per record to simulate user-specific radio and network conditions in an O-RAN system. Each sample includes radio signal indicators such as RSRP, RSRQ, and SINR, as well as QoS-related fields like target throughput, 5QI index, and real-time QoS scores. Resource usage statistics, including PRB consumption, CQI values, and scheduled transport blocks, are provided for both uplink and downlink. Additionally, the dataset contains spatial coordinates ( $x, y, z$ ) and cell-level attributes (e.g., beam ID, slice type), which support the simulation of beamforming, cell association, and access control policies. This dataset serves as the underlying environment for evaluating hybrid action xApp coordination across power control, RB allocation, and beam selection tasks.

The simulation involves 3 base stations serving a total of 6 cells, each operating at a transmit power of 35 dBm. Airspan N5-45x2 Sector Antenna was

selected for the antenna model, since it closely resembles those used in practical 4G/5G O-RAN scenarios, making it well-suited for DT simulations. A total of 30,000 datasets were generated and divided into 60%, 20%, and 20% for the training set, validation set, and testing set, respectively. Each generated cell report contains 41 features and includes data for 20 IoT UEs, each moving at a speed of 1 m/s. To simulate a realistic and complex environment, each UE is randomly assigned a QoS identifier from a total of 23 types of QoS tasks. These QoS tasks are grouped into three major categories: guaranteed bit rate (GBR), non-GBR, and delay-tolerant GBR. The Near-RT RIC is assumed to have two xApps, for power control and resource block, respectively.

## B. BASELINE ALGORITHMS

This section presents the parameter settings used to benchmark multiple RL algorithms in our evaluation. 1. DQN is a foundational value-based RL method, suitable for discrete action spaces such as activating or deactivating specific xApp control policies. In our setup, we configure DQN with a learning rate of  $1 \times 10^{-3}$ , a discount factor of 0.95, and a target network updated every 500 steps. The replay buffer holds  $10^5$  transitions, and the policy network comprises two hidden layers of 128 units each. 2. PPO is a widely used policy gradient algorithm known for its stability. It is applied here as a strong single-agent baseline for isolated xApp decision-making. We use a clipping range of 0.2, learning rate of  $3 \times 10^{-4}$ , and a GAE parameter  $\lambda = 0.95$ . The actor and critic networks each consist of two hidden layers with 64 units. PPO performs well in low-conflict settings but lacks mechanisms to handle inter-xApp dependencies. 3. MAPPO extends PPO to the multi-agent domain under the CTDE paradigm. Each xApp acts as an agent with shared or individual value functions. We set the learning rate to  $5 \times 10^{-4}$  and use a clip range of 0.2 with GAE  $\lambda = 0.95$ . The policy network consists of two 128-unit hidden layers. MAPPO effectively captures cooperative behavior and stabilizes learning in environments with moderate agent interaction, making it well-suited for coordinated xApp deployment. 4. HAPPO improves upon MAPPO by hierarchically updating agent policies in a sequential manner to reduce non-stationarity. In our configuration, we use a Kullback-Leibler (KL) penalty of 0.01, a learning rate of  $3 \times 10^{-4}$ , and batch size of 2048. Agents update their policies one by one while conditioning on others' updates. This makes HAPPO highly effective for scenarios with tightly coupled xApp decisions, such as shared resource management and conflict avoidance. 5. FACMAC incorporates an attention-based centralized critic that dynamically weights the influence of other agents. This mechanism is particularly effective in heterogeneous xApp environments. We configure FACMAC with four attention heads, a learning rate of  $1 \times 10^{-4}$ , and a critic update delay of two steps. The attention network uses 128 and 64 units in its layers. FACMAC is especially valuable for achieving fine-grained policy coordination in dense DT simulations. 6. CommNet enables differentiable inter-agent communication through shared hidden representations. Although simpler than more recent methods, it serves as a conceptual baseline for evaluating the benefits of explicit coordination. Our setup uses a communication vector size of 64, learning rate of  $1 \times 10^{-3}$ , and one to three

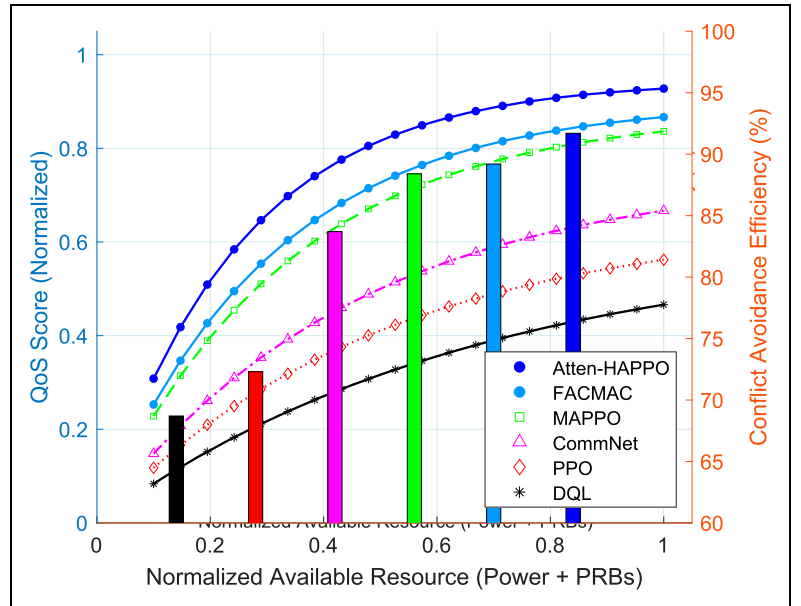


FIG. 5. The CAE and QoS comparison between various reinforcement learning schemes.

communication steps. CommNet is useful for benchmarking how DT-based implicit coordination compares to direct xApp-to-xApp messaging.

## C. CONFLICT AVOIDANCE EFFICIENCY

CAE is a metric designed to evaluate the ability of RL-based xApp management strategies to prevent conflicting control actions in multi-agent O-RAN environments. It quantifies how effectively an RL algorithm can coordinate xApp behaviors to avoid resource contention, policy clashes, or redundant decisions. The CAE is defined as:  $CAE = 1 - \frac{N_{\text{conflict}}}{N_{\text{total}}}$ , where  $N_{\text{conflict}}$  denotes the number of conflict events observed over 10,000 Near-RT RIC control intervals, and  $N_{\text{total}}$  is the total number of such intervals. Each control interval is 10 ms, which corresponds to the lower bound of the control time scale defined by the O-RAN Alliance [15].

To detect conflict events in a multi-xApp O-RAN environment, we consider the following three types of conflict indicators: 1. A direct resource conflict is detected when two or more xApps simultaneously attempt to control the same RAN resource (e.g., PRBs, UEs). 2. A KPI anomaly is flagged when a resource experiences significant performance degradation (e.g., latency spikes or throughput drops) across consecutive control intervals. 3. A policy inconsistency is labeled by the DT simulator when concurrent decisions lead to incompatible control outcomes, such as conflicting handover commands or oversubscription of scheduling blocks.

These mechanisms directly reflect realistic coordination challenges in O-RAN systems. For example, in our evaluation, we simulate a joint control scenario in which one xApp adjusts PRB-based bandwidth allocation while another xApp concurrently manages RU power levels. In poorly coordinated schemes such as DQL and PPO, these xApps independently modify overlapping resources, leading to frequent contention and reduced CAE. In contrast, algorithms like HAPPO and FACMAC enable joint policy learning across agents, allowing them to implicitly resolve such conflicts during training.



| Method  | Training Time | Inference Time | RAM Usage |
|---------|---------------|----------------|-----------|
| HAPPO   | 4.1 hrs       | 1.5 ms         | 3.5 MB    |
| FACMAC  | 3.8 hrs       | 1.6 ms         | 3.2 MB    |
| MAPPO   | 2.7 hrs       | 1.6 ms         | 2.8 MB    |
| CommNet | 3.1 hrs       | 1.7 ms         | 2.6 MB    |
| EC-MARL | 3.5 hrs       | 1.8 ms         | 2.7 MB    |
| IPPO    | 5.4 hrs       | 1.4 ms         | 2.5 MB    |
| PPO     | 2.3 hrs       | 1.3 ms         | 1.8 MB    |
| DQN     | 1.9 hrs       | 1.2 ms         | 1.2 MB    |

TABLE I. Comparison of training and inference complexity across RL methods.

As shown in Fig. 5, HAPPO achieves the highest CAE score of 91.7% by hierarchically updating each agent's policy in a sequence that reduces non-stationarity. FACMAC follows with 89.2%, leveraging attention-based critics to model inter-agent dependencies without explicit messaging. EC-MARL also outperforms single-agent baselines by enabling learned communication among agents, though its lack of global coordination structure limits its scalability.

Overall, the results demonstrate that higher CAE scores are strongly associated with RL algorithms capable of learning coordinated multi-agent strategies—especially in use cases where power control and bandwidth management must operate concurrently on shared resources.

#### D. COORDINATION QoS IMPROVEMENTS

In the previous section, we evaluated the coordination capability of each RL scheme using the CAE metric. However, while CAE effectively captures the degree of coordination among xApps, it does not fully reflect the overall system performance. In this section, we shift our focus to analyzing QoS-related metrics in order to assess the actual performance impact of each RL strategy. CQI is a performance metric used to evaluate the extent to which multiple xApps in an O-RAN system can collaboratively improve overall service quality through coordinated control strategies. It reflects whether reinforcement learning or other coordination mechanisms can effectively enhance KPIs at the network level, such as latency, packet loss, and resource utilization. In this section, we adopt the term QoS Score to consistently represent these improvements across different evaluation scenarios.

From Fig. 5, HAPPO achieves the highest QoS scores, reaching approximately 0.95, by employing a hierarchical policy update scheme that sequentially adjusts each agent's policy, significantly mitigating the non-stationary dynamics commonly seen in decentralized environments. FACMAC follows closely with a QoS score of around 0.90, using an attention-based critic to selectively weigh inter-agent influences. This mechanism enables implicit coordination without explicit communication overhead, offering high adaptability to complex and dynamic O-RAN scenarios. MAPPO provides a stable baseline with a QoS score of 0.88 by leveraging CTDE, but it lacks mechanisms to model agent interdependencies as effectively as HAPPO or FACMAC. CommNet achieves a moderate QoS level of about 0.83 through explicit inter-agent communication using shared embeddings, though it suffers from potential communication noise and limited scalability in

denser deployments. In contrast, PPO and DQN, as single-agent methods, operate independently without awareness of other agents' policies. As a result, PPO reaches only around 0.72 QoS, and DQN further drops to 0.68. This lack of coordination leads to frequent control conflicts and degraded system-level QoS, especially under dense xApp deployments. DQN, being value-based and designed for discrete action spaces, is particularly ill-suited for high-dimensional, resource-sharing environments like O-RAN. EC-MARL incurs moderately higher training time and memory usage than centralized methods due to the added complexity of learning and processing explicit inter-agent communication during training and inference. These results demonstrate that QoS improvements in O-RAN systems strongly correlate with an algorithm's capacity for coordinated decision-making and its robustness against non-stationarity in multi-agent control.

#### E. COMPLEXITY ANALYSIS

As shown in Table I, different RL algorithms exhibit varying computational and memory demands during both training and inference phases. HAPPO and FACMAC, while delivering the highest QoS performance, require comparatively longer training times and higher memory usage due to their multi-agent coordination mechanisms and complex critic structures. MAPPO and CommNet offer a trade-off between performance and efficiency, with moderate training costs and manageable RAM footprints. IPPO incurs the longest training time, as it independently trains each agent without parameter sharing, while PPO and DQN demonstrate the lowest computational costs. EC-MARL incurs moderately higher training time and memory usage than centralized methods due to the added complexity of learning and processing explicit inter-agent communication during training and inference. These results highlight the importance of balancing coordination performance with computational scalability when deploying RL-based xApp management in resource-constrained O-RAN environments.

### IV. CONCLUSION AND FUTURE DIRECTIONS

This paper investigated the use of DT-MARL for intelligent xApp management in IoT-enabled O-RAN systems. By leveraging DT models to simulate large-scale, event-driven IoT behaviors, we demonstrated how proactive scheduling and coordination among xApps can be significantly improved under dense, latency-sensitive scenarios. Comparative evaluations of advanced MARL algorithms, including HAPPO and FACMAC, revealed superior performance in both conflict avoidance and QoS optimization, as measured by two key system-level KPIs. Notably, the proposed MARL approaches maintained acceptable model complexity, supporting practical deployment in resource-constrained environments such as NB-IoT and LTE-M networks. These findings highlight the potential of integrating DTs with decentralized learning frameworks to support scalable, adaptive, and efficient control in next-generation O-RAN deployments. Future work may extend these techniques to online fine-tuning with real-world feedback and integration with standardized O-RAN orchestration interfaces.

Future directions point toward building more agentic AI frameworks for xApp management in O-RAN. These systems emphasize autonomy,

adaptability, and context-aware decision-making. One key challenge is multi-agent credit assignment in dense deployments; HAPPO with role decomposition can help coordinate agents based on their functional roles. In IoT settings with bursty device activity, event-driven MARL can improve responsiveness using asynchronous or gated policy updates. Additionally, agentic RL methods such as Safe DDGP or Lagrangian-based PPO offer a principled way to enforce QoS constraints and tenant fairness in multi-slice environments.

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## BIOGRAPHIES

ZHIZHOU HE (Member, IEEE) received the B.Eng. degree in electrical and electronics engineering from the University of Bristol, U.K., in 2019, the M.Sc. degree in 5G and future generation communication systems from the University of Surrey, U.K., in 2020, and the Ph.D. degree in information and communication systems from the University of Surrey, in 2024. He is currently a Postdoctoral Research Fellow with the 5G Innovation Centre (5GIC) and 6G Innovation Centre (6GIC), Institute for Communication Systems (ICS), University of Surrey. His research interests include ORAN and artificial intelligence.

AHMED AL-TAHMEESSCHI (Member, IEEE) received the Ph.D. degree in electrical engineering and electronics from the University of Liverpool, U.K., in 2018. He served as a Postdoctoral Researcher with Tokyo University of Agriculture and Technology from 2019 to 2021, contributing to 5G-enhance a Horizon 2020 Joint Project. From 2021 to 2023, he served as a Research Associate with the University of Helsinki and was a member of the Finnish Centre of Artificial Intelligence. He is currently a Research Fellow with the University of York, contributing to the YORAN Project. His research interests areas include cognitive radio networks, dynamic spectrum access techniques, algorithm design, localization, open radio access networks, and integration of machine learning into wireless networks.

CHUAN HENG FOH (Senior Member, IEEE) received the M.Sc. degree from Monash University, Melbourne, VIC, Australia, in 1999, and the Ph.D. degree from the University of Melbourne, Melbourne, in 2002. After the Ph.D. degree, he spent six months as a Lecturer with Monash University. From 2002 to 2012, he joined Nanyang Technological University, Singapore, as an Assistant Professor. He is currently a Senior Lecturer with the University of Surrey, Guildford, U.K. His research interests include protocol design, machine learning application and performance analysis of various computer networks including wireless local area networks, mobile ad hoc and sensor networks, vehicular networks, Internet of Things, and 5G/6G networks.

HAMED AHMADI (Senior Member, IEEE) received the Ph.D. degree from the National University of Singapore, in 2012, and has held academic/industry roles in Ireland and U.K. He is a Reader in Digital Engineering, University of York, where he directs the iTWINS Lab and serves as a Deputy Pillar Lead for advanced communications with the Institute for Safe Autonomy. He has authored a book and 100+ peer-reviewed publications, and leads the ML/AI work packages for the Yorkled YO-RAN and REACH 6G O-RAN projects. His interests span ML for wireless, O-RAN, green/sustainable and airborne networks, digital twins, and IoT. He is an Associate Editor-in-Chief of *IEEE Communications Standards Magazine*, a fellow of the UK HEA, and has chaired TPC/OC roles at ICC, EuCNC, and PIMRC.

MOHAMMAD SHOJAFAR (Senior Member, IEEE) received the Ph.D. degree in ICT from Sapienza University of Rome, in 2016. He is an Associate Professor of Network Security with the 5G/6GIC, Institute for Communication Systems, University of Surrey, and an Intel Innovator. His research covers network security, AI/ML for networks, Open RAN, and DT-enabled systems. As PI he has secured ~£2M in EU/UK funding and serves as Associate Editor for *IEEE TRANSACTIONS ON NETWORK AND SERVICE MANAGEMENT*, *IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS*, *IEEE TRANSACTIONS ON GREEN COMMUNICATIONS AND NETWORKING*, and *IEEE TRANSACTIONS ON CONSUMER ELECTRONICS*. He contributes to ETSI ITS/NFV, the GSMA Open-Telco LLM Group, and 3GPP SA, and supported O-RAN Alliance WG11 activities in 2024; he has served on the IEEE Senior Membership Review Panel since 2024.